

Technical Specification – Combat Revision

DATE: 03-OCT-2026 STATUS: DRAFT FOR REVIEW – no code has been changed REVISED: 04-OCT-2026 – incorporates Peter's response ([response-to-feedback-20261003.md](../rules/response-to-feedback-20261003.md)) and the revised default loadout (V2)
IMPLEMENTS: 2026-09-03-combat-revision.md RULES
SOURCES: [fighting-a-match.md](../rules/fighting-a-match.md), [each-round-is-fought.md](../rules/each-round-is-fought.md), [strangers-default-loadout-v2-20261003.md](../rules/strangers-default-loadout-v2-20261003.md) (supersedesstrangers-default-loadout.md), [response-to-feedback-20261003.md](../rules/response-to-feedback-20261003.md), [Expanded Consolidated Rules PV2026 V2.md](../rules/Expanded%20Consolidated%20Rules%20PV2026%20V2.md) (\$FIGHTS, \$Fights between Exploring Parties, ARTEFACTS – WHO CAN USE?)

Open questions – register

Open as of 04-OCT-2026, except the 8 items marked ✓ ANSWERED. There are 45 items: 16 rules questions for Peter (Q-R), 8 default-loadout questions and 3 data discrepancies (Q-L, D), 5 technical questions (Q-T), and 13 product/requirement decisions (G) — **37 still open**. Peter's response of 03-OCT-2026 answered Q-L1 to Q-L5, D1 and D2 (§9.1b) and raised two new items; Mark answered one of them (Q-L7) on 04-OCT-2026. ★ = **blocks the start of M1** (§10). Each item is a short summary; the reasoning, evidence and file references are in the cited section. When an item is answered, record the answer in that section and mark it here.

A. Rules – for Peter (detail: §9.1)

- ★ **Q-R1** — *Triggering a Fight* has no defining document. Who attacks in round 1 for each trigger (hostile reaction, party Attack, pacified-chamber Attack, Demon ambush, retreat bounce-back)? Does the role strictly alternate every round?
 - **Proposed default:** Strangers attack on hostile/Demon; party on Attack; strict alternation
 - **Blocks:** M1
- **Q-R2** — Scroll: is “destroy” usable in any round the party attacks, with the two surprise variants only before round 1?
 - **Proposed default:** Yes
 - **Blocks:** M2
- ★ **Q-R3** — “Front line of any match does not exceed two”: per side, with 2v2 still banned? May unengaged creatures of the larger side join an already-engaged match?
 - **Proposed default:** Per side; 2v2 banned; joining allowed
 - **Blocks:** M1
- ★ **Q-R4** — When and how do magic users commit to the background, for **both** sides? Any number per match? May a lone caster stay back?
 - **Proposed default:** Larger side in step (d), smaller with its front line; a lone caster fights
 - **Blocks:** M1
- ★ **Q-R5** — Stranger casualty: the winning player nominates and on 4–6 the **other** dies — is the nominated creature the one **spared**? Whose Ring adds the +1?
 - **Proposed default:** Nominated = spared; losing side's Ring
 - **Blocks:** M1

- **Q-R6** — Must the Ring-bearer be among the fighters for the side to get +1, or is any living holder enough?
 - **Proposed default:** Any living holder (today)
 - **Blocks:** M2
- **★ Q-R7** — (*Partly answered by the default loadout.*) When is the loadout re-run — only at fight start, or also after a bearer dies, when deserters join, or a Demon arrives later? May strangers pick up artefacts dropped mid-fight?
 - **Proposed default:** Start, and at the next deployment after a bearer dies; no mid-fight pickup
 - **Blocks:** M1
- **Q-R8** — Does a bonus follow the card text (Sword: Man/Woman/Hero only) rather than the *who can use* table? Is the Dwarf-ally Ring blank a typo (the card allows a Dwarf)?
 - **Proposed default:** Card text; typo
 - **Blocks:** M1
- **Q-R9** — Where do Spectre/Demon sit in the new flow, and when does “strongest creature is slain” occur? Can a Spectre be a stranger attacker’s paired creature?
 - **Proposed default:** Resolved at step (e) before matches; engageable only by magic/Sword/Axe/Shield
 - **Blocks:** M2
- **Q-R10** — In a round where the strangers attack, may the player use consumables (Strength Potion, Lotus Dust, Holy Water) or only deploy?
 - **Proposed default:** Only deploy
 - **Blocks:** M2
- **Q-R11** — Hidden Cards variant: may strategies see face-down player cards?
 - **Proposed default:** No
 - **Blocks:** M2
- **Q-R12** — Do casters count toward the defender’s minimum deployment?
 - **Proposed default:** Yes
 - **Blocks:** M1
- **Q-R13** — Multiplayer vs strangers: with several parties or a union, who attacks/defends and in what order?
 - **Proposed default:** One party at a time
 - **Blocks:** M7
- **Q-R14** — *Ending a Fight* has **no document**: end conditions (incl. “put to sleep”), retreat procedure/timing, engaged creatures, loot. Are *Wrap Up the Round* and *Ending a Fight* round/fight-level rather than per-match?
 - **Proposed default:** Today’s behaviour (§4.2.0)
 - **Blocks:** M2
- **Q-R15** — Scope and timing of *Stranger Fight Preparations*: loadout only, or also an initial stance? Once before round 1, or again for reinforcements?
 - **Proposed default:** Loadout only; once, and again after a bearer dies or strangers are added
 - **Blocks:** M1
- **Q-R16** — (*New.*) The loadout document defines how strangers are **equipped**. Is there a matching default **pairing** procedure for the strangers’ deploy / engage / redeploy / background decisions, or should one be derived from the objective in G1?
 - **Proposed default:** Derive from the G1 objective until one is supplied
 - **Blocks:** M2

B. Default loadout and data – for Peter (detail: §9.1a, answers §9.1b)

- ✓ **Q-L1** — ANSWERED. Thief above the Woman in the Sword/Axe lists, a Thief gaining nothing, and a Thief-borne Shield being inert.
 - **Answer (Peter):** The Thief *uses artefacts as a Man*, so it gains what a Man gains; Woman is now ranked **above** Thief (the Unicorn rule makes her ability useful to strangers). The engine does not implement “uses artefacts as” (§4.4.2).
- ✓ **Q-L2** — ANSWERED (values still open: D3). Staff list vs. the engine’s Priest/Wizard-only bonus.
 - **Answer (Peter):** Witch and Scholar use artefacts as a Priest; Apprentice as a Wizard; and, as a **house rule**, the Sorcerer as a Wizard (Q-L7).
- ✓ **Q-L3** — ANSWERED. “Weapon” is no longer used: the lists now say “not bearing Magic Sword” (Axe) and “bearing either Magic Sword or Magic Axe” (Shield).
- ✓ **Q-L4** — ANSWERED. Ring “adjusted strength” = the creature’s strength plus any bonus from bearing the Magic Sword, Magic Axe or Magic Staff.
 - **Still open:** whether the Ring bearer must actually fight (Q-R6).
- ✓ **Q-L5** — ANSWERED. The Eye is **always on** in the area it is in; it need not be borne or held. (Peter: contrast the Talisman, whose card says it has no power until taken up.) The engine’s held-by-the-party test is therefore wrong (§4.4.3).
- **Q-L6** — Confirm the loadout trigger (see Q-R7) and the wording “creature in the party” in Step 5 (the *stranger* party?).
 - **Proposed default:** Confirm
 - **Blocks:** M0
- ✓ **Q-L7** — ANSWERED. **Sorcerer uses artefacts as a Wizard** (house rule): Consolidated Rules text, or Addendum?
 - **Answer (Mark, 04-OCT-2026): Addendum**, referenced from the main text where relevant. Recorded as note #4 in Expanded Consolidated Rules PV2026 V2.md, marked as not original, with references from the Sorcerer, Magic Staff and The Ring entries and the who-can-use table. The engine and this spec treat it as a rule, flagged as a house rule.
- **Q-L8** — (*New.*) The Woman-Hero’s card reads “all the capabilities of a woman and a hero”, not “uses artefacts as a Hero”. Confirm the union reading: she may bear and gain what either a Woman or a Hero would.
 - **Proposed default:** Union of Woman and Hero capabilities
 - **Blocks:** M0
- **D3** — Data: the Staff card says +2 for Priest **or** Wizard; the engine gives the Priest +1. Deliberate house rule? (Not addressed in Peter’s response.)
 - **Proposed default:** Peter to decide
 - **Blocks:** M0

- ✓ **D1** — ANSWERED. Hero STR is **5**: Peter’s revised Consolidated Rules (Consolidated–Rules–20260925–2.pdf) say 5 and the engine already has 5. The old text’s 6 was a transcription slip.
- ✓ **D2** — ANSWERED. Spectre is **STR 5 (MAG 5)** in the revised rules; the engine’s Spectre magic is already 5 (its fighting strength 0 reflects “cannot be fought hand-to-hand”).

C. Technical – for engineering (detail: §9.2)

- **Q-T1** — Is `combatRevision` a per-game flag that defaults off (this spec’s assumption), or do we migrate everything and retire the old engine?
 - **Proposed default:** Per-game flag; flip the default later
 - **Blocks:** M1
- **Q-T2** — Neural network: hosting, auth, latency SLA, cost ceiling — and may we send full state or only the human-visible view?
 - **Proposed default:** Human-visible view only
 - **Blocks:** M6
- **Q-T3** — What CPU budget can the expert strategy use inside a Convex mutation? (Measure against current limits.)
 - **Proposed default:** Measure first
 - **Blocks:** M5
- **Q-T4** — Should PvP migrate onto the new round engine, or stay on `multi-fight.ts`?
 - **Proposed default:** Migrate in M7
 - **Blocks:** M7
- **Q-T5** — May we add a property-testing dependency (`fast-check`)?
 - **Proposed default:** Yes
 - **Blocks:** M1

D. Product and requirement decisions (detail: §9.3)

- **G1** — Define “maximum effect” and what the difficulty levels mean.
 - **Proposed default:** §5.4 objective and levels
 - **Blocks:** M2
- **G2** — The neural-network contract: model, inputs, SLA, hosting, cost, fallback, privacy, versioning, ownership.
 - **Proposed default:** Draft in §5.6
 - **Blocks:** M6
- **G3** — What may a strategy see (hidden cards, other parties’ secrets)?
 - **Proposed default:** Human-visible only
 - **Blocks:** M2
- **G4** — Backwards compatibility for in-flight games, saved codes, replays and golden snapshots.
 - **Proposed default:** Per-game flag (Q-T1)
 - **Blocks:** M1
- **G5** — Multiplayer scope: PvE only, or PvP too? Do 45-second windows apply to stranger decisions?
 - **Proposed default:** PvE first; no windows
 - **Blocks:** M7
- **G6** — UX for *defending* and for step (d): wireframes, mobile, accessibility.
 - **Proposed default:** Draft in §7.1
 - **Blocks:** M4

- **G7** — Should leaderboards be segmented by difficulty level (as they are by kit)?
 - **Proposed default:** Yes
 - **Blocks:** M5
- **G8** — How are stranger decisions logged and shown in replay, and how does a bug report show “why”?
 - **Proposed default:** §5.2 logged actions + why trace
 - **Blocks:** M3
- **G9** — Test Mode support: force a level, script a stranger formation.
 - **Proposed default:** Add controls
 - **Blocks:** M3
- **G10** — Success metrics: win-rate bands per level, decision latency, fallback rate.
 - **Proposed default:** Set after the bench (§8)
 - **Blocks:** M5
- **G11** — Ship R5 (UI legality) first on the current engine, as M4a?
 - **Proposed default:** Yes
 - **Blocks:** M4a
- **G12** — One rules authority: how Peter’s documents trace to the engine spec.
 - **Proposed default:** Appendix A
 - **Blocks:** M0
- **G13** — *Triggering a Fight* and *Ending a Fight* have no rules documents; overview and Peter’s step names differ.
 - **Proposed default:** §4.2.0 mapping
 - **Blocks:** M0

0. Summary

Peter’s two rules documents describe combat as a **round-by-round engagement protocol**: the attacker and defender *alternate each round*; the defender deploys a front line; the attacker pairs against it; the larger side then redeploys within limits; matches that are unresolved **persist** into the next round; and strangers use artefacts “to best advantage”.

The current implementation is a different shape. The *player* supplies a complete pairing every round, strangers are an undifferentiated list of creature ids with no equipment and no decisions, nothing persists between rounds, and the stranger side is never an attacker. Implementing the revision therefore means:

1. a **new round engine** (phases a–f) with persistent engagements and alternating roles,
2. **stranger inventories** (so strangers can bear artefacts),
3. a **pluggable decision layer** for everything the stranger side must decide, with stranger decisions recorded as ordinary logged actions (so replay never calls a strategy),
4. an **engine-owned legality layer** that both the UI and the strategies use, so an illegal formation cannot be built, and
5. a **strategy test harness** that can evaluate strategies exhaustively.

All of it ships behind a per-game rules flag (`variants.combatRevision`) so in-flight games, saved codes, replays and golden snapshots keep their current behaviour.

Two engine corrections from Peter’s response are also needed, and are part of M0 (§4.4.2, §4.4.3): a **capability table** so “uses artefacts as” is honoured (a Thief, Witch, Scholar or Apprentice currently gets no

bonus from artefacts a Man, Priest or Wizard would), and an **area-wide Eye of God** (today only an Eye held by the party counts).

The engine phases follow the **Combat Round Overview** in the high-level spec one-to-one (§4.2.0). Two of its steps — *Triggering a Fight* and *Ending a Fight* — are not defined by any rules document yet (Q-R1, Q-R14).

The questions that block the start of M1 are marked ★ in the register above; everything else is tracked there too.

1. Current state (what the code does today)

1.1 Engine – solitaire and multiplayer-vs-strangers (packages/engine/src)

Aspect	Today	Where
Strangers	<code>state.strangers: number[]</code> — creature ids only. No equipment, no per-stranger state. Chamber draw is capped at <code>MAX_STRANGERS = 8</code> , but a Mutiny adds every ally as a stranger with no cap (WAMV reached 14; theoretical ceilings 37 standard / 48 extended, neural-net feasibility study §2.6).	<code>state.ts</code> , <code>chamber.ts:6</code>
Who pairs	The player sends <code>resolveRound { matches: PlanMatch[] }</code> every round: per match <code>front[1–2]</code> , <code>backers[]</code> , <code>strangers[1–2]</code> .	<code>actions.ts:20</code> , <code>combatPlan.ts</code>
Stranger “strategy”	None. When the party runs out of free fighters the engine auto-attaches the strongest leftover strangers onto lone fighters (“strongest combination”, §395) and folds leftover enemy casters’ MP into the first non-Spectre match.	<code>previewPlan</code> , SC-9.1-10/11
Roles	No attacker/defender concept. Round 1’s surprise (+1 party, –1 strangers) is the only trace of who initiated.	<code>FightState</code>
Persistence	None. A <code>BattlePlan</code> is rebuilt every round. There is no “engaged” state.	<code>state.ts:104–123</code>

Aspect	Today	Where
Strength	frontStrength, casterMP, enemyMP, partyRollBonus — party and stranger arithmetic live in different functions (combat.ts vs combatPlan.ts). Artefact bonuses are possession-based for the party and absent for strangers.	combat.ts, combatPlan.ts:previewPlan
Casualties	One loser → dies. Two front-line losers → casualtyQueue; player names a preference; d6 4–6 grants it. Stranger casualties: strongest foe in the match dies, automatically.	SC-9.5-3/4/5, SC-9.3-10
Scroll	One variant only (destroy every non-magical stranger), offered in encounter and fight in any round, then curses.	reduce.ts:1447
Immaterial foes	Spectre/Demon rules are spread through validation, preview and resolution (MAGIC_ONLY_IDS, magicOnlyBypass, forced-Spectre round).	combatPlan.ts
Retreat	Legal only in fight, round > 1, not after a trap fall; dead end sets retreatBlocked.	SC-9.5-6/7

Fight starts: hostile reaction / hostile-on-sight → startFight(state, -1); Demon ambush → -1; Attack with a fresh entry → +1; Attack on a pacified chamber → 0 (reduce.ts:147–164, 1113, 1129, 1134).

1.2 Engine – player-versus-player (multi-fight.ts, ~750 lines)

PvP is a **separate implementation** that already models the shape Peter describes: declarePvp → setDefenderLine → setAttackerEngage → setDefenderCasters → resolveRoundPvp, with PvpEngagement { attackers, defenders }, PvpBacker, an alternating round owner, 45-second reaction windows, and per-side dice substreams. It reuses solo arithmetic (frontStrength, casterMP) but not the solo pairing. **Peter’s rules are, in effect, the PvP protocol generalised to stranger parties.**

1.3 Server (apps/web/convex)

- game.applyAction is a Convex **mutation**: auth → reduce(state, action) → patch games.state → append {seq, action, events} to gameEvents. multiplayer.ts has its own equivalent; multiplayer fights against strangers go through the same pure reduce via compose(mp, party) (multi.ts:516).

- **Replay** is `replay(seed, picks, actions, variants, testMode)` — a pure fold of the logged actions through `reduce`. A game’s correctness on replay depends on `reduce` being deterministic given the log.
- The action argument validator is **hand-maintained and duplicated** in `game.ts:14` and `multiplayer.ts:~20` (a past regression is recorded at `game.test.ts:515`). Any new action or field must be added to both.
- `variants` is a typed object on the games row (`schema.ts`), threaded through `newGame`, `startTestGame`, `replay`, `replayByCode`, `highScores.log` and the web `ReplayBundle`.

1.4 Front end (apps/web/src/game)

- `FightSurface.tsx` (355 lines) builds a `PlanDraft` (`fightPlan.ts`, 60 lines) by tap/drag; it calls `validatePlan` and `previewPlan` **after** every placement and shows a reason string. Illegal drafts can be built (`place` allows a backer with no front fighter; the plan is merely flagged).
- Rule knowledge leaks into the UI: `REASON` strings, `MAGIC_ONLY` ids (`C_SPECTRE`, `C_DEMON`), the forced-Spectre derivation and the doomed-member calculation are all re-derived in the component.
- `PvpFightSurface.tsx` (493 lines) is a second, structurally different fight UI.

1.5 Gap analysis – Peter’s steps against today

Peter	Rule	Today	Gap
(a)	Scroll: attacker only, not under the Eye; 3 variants (destroy non-magical / double surprise / add surprise); then cursed	One variant; any round; no attacker check	Variants 2 and 3 missing; “attacking only” not enforced; Eye check not seen in the Scroll path (to verify)
(b)	Defender deploys $\geq$ as many unengaged creatures as the attacker has unengaged; may hold surplus back	No defender step	New phase and new decision
(c)	Attacker pairs its unengaged creatures one-to-one against the defender’s unengaged front line	Player pairs freely; strangers never pair	Strangers cannot attack; pairing is not “against a deployed line”
(d)	Larger party redeploys unengaged + background creatures; engaged creatures fixed; number of matches fixed; ≤ 2 front-line per match	Gang-up is automatic and only when the party is out of fighters	The larger side’s redeployment is a real decision (and a strategy hook for strangers)
(e)			

Peter	Rule	Today	Gap
	Fight each match (Step i–iv)	Implemented, but strength arithmetic is split	Unify; add stranger artefacts, Shield match-scoping, Ring and surprise $\{0, 1, 2\}$
(f)	Dissolve dead matches; background → unengaged; engaged front line stays; alternate turn	No persistence; no alternation	New state and new flow
Step ii–iv	Stalemate (both zero) / tie / win; tie and stalemate are unresolved	tie → nothing happens (no persistence)	Persist unresolved matches
Casualty	Stranger casualty choice passes to the winning player ; d6 4–6 → the other creature dies	Strongest foe dies automatically	New stranger-casualty rule (see Q-R5)
Overview: <i>Triggering a Fight</i>	A named step; no rules document defines it	Five startFight call sites choose only a surprise value (+1 / 0 / –1)	No explicit attacker/defender, no trigger kinds (Q-R1)
Overview: <i>Stranger Fight Preparations (stranger party only)</i>	Artefact allocation per strangers–default–loadout–v2–20261003.md; possibly more	Nothing — strangers carry nothing	New phase before round 1 (§4.2.0, §4.4.1); scope to confirm (Q-R15)
Overview: <i>Ending a Fight</i>	Referenced by Peter’s step (f) (“See: Ending a Fight”) but no document exists	Rules scattered in SC-9.5 (retreat, loot, wipe-out) and finalizeRound	A defined end-of-fight procedure is missing (Q-R14)

2. Goals and non-goals

Req	Goal	Delivered by
R1	Pair strangers for maximum effect	§5 strategies; §5.4 baseline objective
R2	Strangers use artefacts	§4.4 stranger inventory; §4.3 effective-strength engine
R3	Player must match strangers as best they can	§4.2 round engine; §4.5 legality layer (mandatory deployment)

Req	Goal	Delivered by
R4	Pluggable pairing strategies	§5.1 interface; §5.5 registry/levels; §5.6 external strategy
R5	UI guarantees a legal formation	§4.5 legality layer; §7 front end
Testing	Exhaustive strategy evaluation at scale	§8

Non-goals (this spec): changing the reaction/encounter rules; Hidden Cards and Zombies variants (flagged in §9); the content of the neural network itself (only its integration contract); re-theming the fight UI beyond what the new phases require.

3. Design principles

1. **One implementation.** Solo, multiplayer-vs-strangers and (later) PvP run on the same round engine. No new rule is written twice.
 2. **Pure and deterministic.** reduce stays synchronous and replayable. Anything non-deterministic or remote (a neural network) happens **outside** reduce and enters as a logged action.
 3. **Stranger decisions are first-class actions.** They are validated by the engine exactly like player actions, written to gameEvents, and shown in replays. A strategy is therefore never needed to replay a game.
 4. **Legality is single-sourced.** The engine exposes the legal moves; the UI and every strategy consume them. Nothing re-derives a rule.
 5. **Versioned rules.** variants.combatRevision selects the new engine per game. Old games are untouched.
 6. **Strategies see what a human would see** (all cards face-up unless a Hidden Cards variant is active).
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4. Engine design

New module tree under packages/engine/src/combat/ (the legacy combatPlan.ts / combat.ts stay for combatRevision: false and are later thinned to adapters over the new strength module).

```
combat/
  model.ts      Combatant, Match, Formation, FightV2, StrangerInstance, decisions
  artefacts.ts  data-driven "who can bear / what it does" table (Peter's matrix)
  strength.ts   Step i-iv: effectiveStrength, sideTotal, result, outcome
  legality.ts   legal moves + formation validation (single source for UI and strategies)
  round.ts     the a-f phase machine (pure transitions + events)
  casualty.ts   party and stranger casualty rules
  strategy/    interface, registry, built-ins (novice / standard / expert), evaluator
  index.ts
```

4.1 Domain model

```
export type Side = "party" | "strangers";
```

```
/** A creature in the fight, addressable independently of its array position. */
export interface CombatantRef { side: Side; idx: number } // idx into state.party /
state.strangerSet
```

```

export interface StrangerInstance {
  creatureId: number;
  carrying: number[];           // treasure ids this stranger holds (artefacts it may bear)
  borne: number[];             // subset currently borne ("to best advantage")
}

/** One engagement. Each side has 1-2 front-line creatures and any number of background casters.
 */
export interface Match {
  id: number;                   // stable across rounds so logs/UI can follow a match
  front: { party: number[]; strangers: number[] };
  back: { party: number[]; strangers: number[] }; // casters lending MAG; reset each round (Step f)
}

export interface FightV2 {
  trigger: FightTrigger;        // how the fight began – fixes the first attacker and the surprise bonus (Q-R1)
  stage: "prep" | "rounds" | "ended"; // Stranger Fight Preparations → rounds → Ending a Fight
  round: number;
  attacker: Side;               // alternates each round (Q-R1)
  firstRoundSurprise: { party: 0 | 1 | 2; strangers: 0 | 1 | 2 }; // Scroll can raise it (Q-R2)
  matches: Match[];             // persisted engagements (unresolved matches carry over)
  deployed: { party: number[]; strangers: number[] }; // unengaged creatures already in the front line this round
  pending: PendingDecision | null; // who must decide what next (drives UI and drivers)
  scrollUsedHere: boolean;       // "cannot be used twice in one area"
  casualtyQueue?: CasualtyChoice[];
  retreatBlocked?: boolean;
}

export type PendingDecision =
  | { by: Side; kind: "scroll" } // attacker only
  | { by: Side; kind: "deploy"; min: number } // defender, step (b)
  | { by: Side; kind: "engage" } // attacker, step (c)
  | { by: Side; kind: "redeploy" } // larger side, step (d)
  | { by: Side; kind: "casualty"; pair: CombatantRef[] };

```

engaged is derived (a creature is engaged iff it is in the front line of a match), never stored.

State change. `state.strangers: number[]` becomes `state.strangerSet: StrangerInstance[]` when `combatRevision` is on. Everything that reads `state.strangers` (reaction, hazards, Mutiny deserters, Scroll, Holy Water, Lotus Dust, Gallery statues, projection/rendering) must go through a small accessor (`strangerIds(state)`) during the transition. This is the largest blast radius in the change; see §10 risks.

4.2 Round engine (Peter's steps a-f)

4.2.0 Fight lifecycle – the Combat Round Overview

The phases below are the **Combat Round Overview** of the high-level spec, named identically so the document and the code can be read side by side.

Overview step	Peter's source	Engine phase	Decided by	Notes
Triggering a Fight	<i>none</i>	trigger	engine	Sets <code>trigger</code> , the first attacker and the surprise bonus (below)
Stranger Fight Preparations (stranger party only)	strangers-default-loadout-v2-20261003.md	prep	strangers' <code>assignArtefacts</code>	Once, before round 1; runs <code>defaultLoadout</code> (§4.4.1)
The Scroll Option	Each Round, step a	round.a	attacker	
Defender Prepares Their Defense	step b	round.b	defender	
Attacker Creates Engagements (Matches)	step c	round.c	attacker	
Deployment is Finalised	step d	round.d	larger party	
Calculate Effective Strength per Creature	Fighting a Match, step i	match.i	engine	<code>strength.effectiveStrength</code>
Calculate Total Effective Strength Per Side	step ii	match.ii	engine	<code>strength.sideTotal</code>
Roll Dice & Determine Results	step iii	match.iii	engine	one d6 per side; surprise {0,1,2}, Ring {0,1}
Match is Resolved	step iv	match.iv	engine	stalemate / tie → unresolved; win → resolved or casualty pair
Wrap Up the Round	Each Round, step f	round.f	engine	
Ending a Fight	referenced by step f; no document	end	engine	Q-R14

The overview lists *Wrap Up the Round* and *Ending a Fight* under “Each Match in the Round is Fought”. I have treated them as **round-level** and **fight-level** steps, as in Peter's step (f); please confirm (Q-R14).

Triggering a Fight. Today five startFight call sites choose only a surprise value (Demon ambush -1, strangers' surprise -1 ×2, party Attack +1/0). The revision makes the trigger explicit:

```
export type FightTrigger =
| { kind: "partyAttack";      attacker: "party";      surprise: 0 | 1 | 2 }    // fresh
  doorway/carpet entry → 1; pacified chamber → 0
| { kind: "strangersAttack";  attacker: "strangers"; surprise: 1 }    // hostile
  reaction / hostile-on-sight
| { kind: "demonAmbush";      attacker: "strangers"; surprise: 1 }    // kit
| { kind: "retreatBounceBack"; attacker: "strangers"; surprise: 0 };  // dead-end
  retreat – fight another round (Q-R1)
```

The surprise value is the *first-round* bonus for the **attacking** side; Scroll variants can raise it (Q-R2).

Ending a Fight (provisional). Until Peter supplies the missing rules, end reproduces today's behaviour so nothing regresses: the fight ends when one side has no creature able to fight (party wiped → gameOver; strangers cleared → fightWon, floor treasure reclaimed, fallen creatures' carried items swept into the pickup — SC-9.5-9/11), or when the party retreats successfully (strangers and floor treasure left behind, area becomes hostile — SC-9.5-6/7/8). Strangers never retreat.

Each step is a pure function (state, decision) → { state, events }; round.ts owns the sequencing and sets fight.pending to say who acts next.

Step	Phase	Who decides	Engine action	Notes
a	Scroll option	Attacker (if able and no Eye and not yet used here)	useScroll { variant } or skipScroll	Variants per Peter; cursed afterwards; strangers cannot read it
b	Defender deploys	Defender	deploy { creatures[] }	Must deploy ≥ the attacker's unengaged count (or all it has); may hold surplus back
c	Attacker engages	Attacker	engage { pairs[] }	One-to-one against the defender's <i>unengaged front line</i> until all engaged or none possible
d	Finalise deployment	The larger party	redploy { placements[] }	Engaged creatures fixed; unengaged and background may move; match count fixed; ≤ 2 per side per

Step	Phase	Who decides	Engine action	Notes
				match; 2v2 banned (Q-R3)
e	Fight	— (engine)	automatic	Per match: Steps i–iv; unresolved → persists; casualties queued
f	Wrap up	— (engine)	automatic	Dissolve matches with an empty front line; background → unengaged; flip attacker; open retreat for the party

Action names here are the engine vocabulary; the Convex validators change accordingly (§6).

Alternation in solitaire. A “round” ends a turn (Peter / consolidated rules). Round n odd/even determines who attacks. Rounds where the *strangers* attack have the player as **defender**: the player’s decisions are deploy (b) and, if the party is the larger side, redeploy (d). Rounds where the party attacks have the **strangers** deploy first and the player engages (c). This is the first time the stranger side makes a decision in solitaire.

Strength (Step i–iv) is computed by `strength.ts` for *both* sides with the same code:

```
effectiveStrength(c: Combatant, ctx: MatchContext): number // positional + artefact +
    immaterial effects
sideTotal(match, side, ctx): number // TES
result(match, side, roll): number // TES + d6 + surprise{0,1,2} +
    ring{0,1}
outcome(a, b): "stalemate" | "tie" | { winner:
    Side } // stalemate: both zero; tie: equal non-zero
```

`ctx` carries the Eye, curses, level (Ring invincibility), the Shield’s match-scoped nullification, and the immaterial-creature rules. The artefact effects are **data**, not code branches (§4.3).

Outcome handling: - stalemate / tie → match **unresolved** and persists (a Shield-vs-Spectre standoff is a stalemate, as today). - win, loser front line = 1 → that creature dies; match **resolved**. - win, loser front line = 2 → casualty pair queued; match **unresolved** (§4.6).

4.3 Artefact effects as data

`artefacts.ts` encodes Peter’s *WHO CAN USE?* matrix and each card’s effect once:

```
interface ArtefactRule {
  treasureId: number;
  canBear(creatureId: number, role: "ally" | "stranger"): boolean; // the matrix (incl.
    house-rule note #3)
  effect(creatureId: number, ctx: MatchContext):
    Effect[]; // strength +n, die +n, nullify MAG, enables-fight-Spectre/
    Demon, ...
```

```

    nullifiedByEye: boolean;
}

```

Matrix as parsed from the rules document (stranger columns are new engine behaviour):

Artefact	Human ally	Priest-class ally	Inhuman ally	Dwarf ally	Human stranger	Priest-class stranger	Inhuman stranger	Dwarf stranger
Magic Staff	–	yes (#1)	–	–	–	yes	–	–
Magic Sword	yes	yes	–	–	yes	yes	–	–
The Ring	yes	yes	–	–	yes	yes	–	yes
Magic Axe	yes	yes	–	yes (#2)	yes	yes	–	yes (#2)
Magic Shield	yes	–	–	–	yes	–	–	–

(Consumables — Strength Potion, Lotus Dust, Holy Water, Scroll, Talisman, Flute, Carpet — remain party-only; strangers use none.)

Able to bear is not the same as *gets a bonus*: e.g. the Sword’s strength bonus applies only to Man/Woman/Hero (+1/+2), so a Priest-class stranger bearing it gains nothing from it (Q-R8). The rule table must carry both.

This replaces the possession-based, party-only logic in `frontStrength` / `casterMP` / `previewPlan`, including the duplicated “modifier chip” arithmetic in `previewPlan`.

4.4 Stranger artefacts

- **Pool.** Artefacts available to strangers at fight start = artefacts in the chamber’s guarded treasure (`state.treasures`) plus anything carried by Mutiny deserters (they already drop loot into `treasures`). Q-R7 asks Peter to confirm, and whether dropped party artefacts become available mid-fight.
- **Assignment.** The `assignArtefacts` hook (§5.1) decides who bears what, validated against `ArtefactRule.canBear`. The default implementation is Peter’s procedure, §4.4.1. It runs once at fight start (re-run policy: Q-R7). The party cannot redistribute during a fight today (`PartyPanel` note), so the same asymmetry is kept unless Peter says otherwise.
- **Death.** A dying bearer drops its artefacts to the floor (`state.treasures`), same as party corpses (`sweepFallen`). Scroll-destroyed strangers also drop what they carry (Peter’s note); an invulnerable Ring-bearer who is defeated on level ≥ 4 **disappears with the Ring** and leaves other items.
- **Display.** Cards on the fight surface show a stranger’s borne artefacts exactly as the party’s do (`FightCard`’s relic corner already supports it).

4.4.1 Default loadout (strangers-default-loadout-v2-20261003.md)

Run in the **Stranger Fight Preparations** phase (§4.2.0), this is a pure function `defaultLoadout(strangers, present, ctx, rng) → ArtefactAssignment`, implemented as **data plus a tiny interpreter** (the preference lists are tables, not code), so Peter can change an order without touching logic.

- **Eye present** → **skip**. All other fight artefacts are powerless, so none is allocated and everything stays on the floor. “Present” means **in the area**, held or not (Q-L5, §4.4.3) — the engine does not test that today.
- **Who may bear what** is decided by capability class, not creature id (§4.4.2): the lists below name creature types, but a Thief counts as a Man, a Witch or Scholar as a Priest, an Apprentice as a Wizard, and the Sorcerer as a Wizard (house rule, Addendum note #4).
- **Strictly sequential**, each step seeing the earlier steps’ results:

Step	Artefact	Preference order (first eligible wins)
1	Magic Sword	Hero, Woman-Hero, Man, Woman, Thief
2	Magic Axe	Dwarf; then, not bearing the Magic Sword : Hero, Woman-Hero, Man, Woman, Thief
3	Magic Staff	Sorcerer, Apprentice, Wizard, Witch, Priest, Scholar
4	Magic Shield	bearing the Sword or the Axe : Hero, Woman-Hero, Man, Woman, Thief; then bearing neither : the same five
5	The Ring	Highest <i>adjusted strength</i> (the creature’s strength plus any bonus from bearing the Sword, Axe or Staff) among those able to bear it; ties by Sorcerer, Apprentice, Wizard, Hero, Witch, Woman-Hero, Priest, Man, Scholar, Woman, Thief , Dwarf

- *Transcription note*. V2 of the document reads “Man already not bearing...” at Shield item 8; this spec reads it as “Man not bearing...”. The word “weapon” no longer appears, so Q-L3 is closed.
- **Ties**. Equal eligibility → random choice; equal type and equal eligibility → random choice. The randomness comes from the decision’s `rngSeed` substream (§5.1) — never the game seed — and the chosen assignment is written to the log as a `strangersAssignArtefacts` action, so replay reproduces it without re-rolling.
- **Unbearable** → **floor**. An artefact nobody can bear stays in `state.treasures`.
- **Allocated** → **carried**. An allocated artefact moves from the floor into that stranger’s carrying/borne.

- **One creature may bear several artefacts** (e.g. Hero with Sword, Shield and Ring); the Axe/Sword “only one used per fight” rule is applied by strength, not by the loadout. Test vectors for this procedure are in Appendix C.

4.4.2 Capability resolution – “uses artefacts as” and “has all the capabilities of” (Peter’s response)

Artefact cards were printed for the base game, before the Thief, Witch, Scholar and Apprentice existed, so *who may use an artefact* cannot be read from the artefact card alone: the creature card must be consulted too. Peter’s rulings, to be implemented as **one table** (capabilities) read by the loadout, strength, casterMP, the Shield ward and the Spectre/Demon predicates, instead of hard-coded creature ids:

Creature	Counts as	Source
Thief	Man (“uses artefacts as a Man”)	creature card
Witch, Scholar	Priest (“uses artefacts as a Priest”)	creature card
Apprentice	Wizard (“uses artefacts as a Wizard”)	creature card
Woman-Hero	Woman and Hero (“all the capabilities of a woman and a hero”)	booklet text (Q-L8)
Sorcerer	Wizard — house rule , absent from the original rules; recorded as Addendum note #4 (Q-L7)	Peter’s response; Mark’s decision 04-OCT

What the engine does today (checked against packages/engine/src):

- **Hard-coded id lists.** The Sword bonus (combat.ts:33), the Axe bonus, the Spectre-with-Sword predicate (combatPlan.ts:36) and the Shield ward (SHIELD_WARD_ELIGIBLE, effects.ts:13) all test creatureId against [0, 1, 5, 6] (Hero, Woman-Hero, Man, Woman). A **Thief gets no Sword/Axe bonus and an inert Shield**.
- **Staff.** casterMP (combat.ts:55) adds +1 only for creature 4 (Priest) and +2 only for creature 8 (Wizard), so the Witch, Scholar, Apprentice and Sorcerer gain nothing from a Staff they bear.
- **usesArtefactsAs exists** (effects.ts) and is applied for the *non-fight* artefacts (Healing Balm, Flute, Carpet, the Staff’s revive and Medusa-ward, selectors.ts, hazards.ts:45) but **not** in the strength code above.

Peter’s suspicion of a systemic bug is confirmed. Consequences for the plan:

1. **Behaviour change under the current rules.** Fixing it alters fight results for Thief, Witch, Scholar and Apprentice parties (and, with the Sorcerer house rule, a Sorcerer). It is therefore **flag-gated with combatRevision** (Q-T1) so existing golden snapshots are untouched; a separate decision is needed on whether to also fix the legacy path.
2. **Mutual dependency.** The loadout and the strength calculation must use the same capability table, or a stranger could be handed an artefact it cannot use (the Thief-with-Shield case).
3. **MO work item.** Capability table, the tests that pin it, and the corrected strength/ward/predicate call sites.

4.4.3 The Eye of God is area-wide, not carried

Peter reads the card (“powerless **when this gem is in the same area**”) against the Talisman (“no warding-off power until actually taken up”) and concludes the Eye is **always on in the area it is in**; a party need not pick it up to suppress magic and artefacts, and one that has taken it must keep it with the main body or be cursed. Therefore `eyePresent(state)` must be true when the Eye is **held by a party member or lying in the current area’s floor treasure**, and it applies to **both sides** (strangers’ loadout skipped; the player’s artefacts and magic also nullified). Today `eyeActive(effects.ts:90)` is true only when a party member holds it, so an Eye left on the floor does nothing; this is an M0 engine correction, gated like §4.4.2. An Eye on the floor in an *adjacent* area has no effect. (Whether a party that declines the Eye should have the Scroll blocked too is covered by the existing Eye check in Step a.)

4.5 Legality layer (R3, R5)

`legality.ts` is the **only** place formation rules live. It is used by the engine to validate, by strategies to enumerate, and by the UI to constrain.

```
legalDeployments(state, side): { min: number; candidates: CombatantRef[] }           // step b
legalEngagements(state, side): { open: CombatantRef[]; targets: CombatantRef[] }     // step c
legalPlacements(state, side, draft, creature): Placement[]                         // step d,
    per creature
validateFormation(state, side, draft): { ok: true } | { ok: false; reasons: Reason[] }
completeFormation(state, side, draft): Formation // fills the mandatory remainder
    deterministically
mandatory(state, side): Obligation[] // e.g. "must deploy 3", "must engage the
    Ogre"
```

Properties (all property-tested, §8): - **Closure**: `legalPlacements` never returns a placement after which `validateFormation` could fail *unless* the draft is still incomplete; `completeFormation(draft)` always yields a valid formation. - **Totality**: if any legal formation exists, `completeFormation` returns one; if none exists the engine reports the forced outcome (the Spectre/Demon “strongest creature is slain” rule becomes an explicit obligation). - The existing rules (backer must be a caster by creature type, a backer needs a front fighter, 2v2 banned, Spectre/Demon need magic or Sword/Axe/Shield, must-engage-all) are re-expressed here and the legacy `validatePlan` becomes a thin adapter for `combatRevision: false`.

4.6 Casualties (new)

- **Party loses a 1-creature front line**: that creature dies (unchanged).
 - **Party loses a 2-creature front line**: the party nominates one; d6 (+1 if the losing party has the Ring, 7 counts as 6): 4–6 → the nominated creature dies; 1–3 → the other (unchanged, SC-9.5-5).
 - **Strangers lose a 2-creature front line**: the choice passes to the **winning player**; d6 4–6 → the **other** (not nominated) stranger dies. *As written this reads inverted relative to the party rule; Q-R5 asks for the intended meaning before it is coded.*
 - The small card of the dead remains in the area (Healing Balm window — existing `diedTurn/diedArea`).
 - A dead Sorcerer, Dragon and so on keep their existing hooks (`sorcererSlain`, `dragonSlain`).
-

5. Pluggable stranger strategies (R1, R4)

5.1 Interface

Strategies decide only what a human opponent would decide. There is **no casualty hook** (the winning player chooses stranger casualties) and **no Scroll hook** (strangers cannot read it).

```
export interface StrangerStrategy {
  readonly id: string;           // "novice" | "standard" | "expert" | "nn-v1" | ...
  readonly level: StrangerLevel;
  readonly kind: "builtin" | "external";

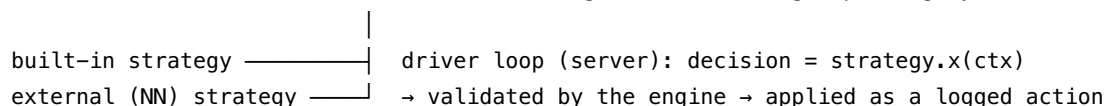
  assignArtefacts(ctx: StrategyContext): ArtefactAssignment | Promise<ArtefactAssignment>; //
    default = §4.4.1
  deploy(ctx: DeployContext): DeployDecision | Promise<DeployDecision>;           // step b
    (strangers defending)
  engage(ctx: EngageContext): EngageDecision | Promise<EngageDecision>;           // step c
    (strangers attacking)
  redeploy(ctx: RedeployContext): RedeployDecision | Promise<RedeployDecision>; // step d
    (strangers the larger side)
}

interface StrategyContext {
  view: FightView;           // read-only projection: both sides' creatures, borne artefacts,
    matches, round, level, Eye, curses
  legal: LegalMoves;         // from legality.ts – the ONLY source of what is allowed
  evaluate: Evaluator;        // exact expected-outcome helpers (§5.3)
  budget: { maxNodes: number; deadlineMs?: number };
  rngSeed: number;           // a per-decision substream; strategies never touch the game seed
}
```

Decision objects are plain, serialisable data (they become logged actions — §5.2).

5.2 Where strategies run, and how decisions are logged

player action → reduce() → state awaits a stranger decision (fight.pending.by === "strangers")



- **Stranger decisions are actions:** strangersDeploy, strangersEngage, strangersRedeploy, strangersAssignArtefacts, each carrying { strategyId, strategyVersion, decision }. They go through reduce and are appended to gameEvents like any player action.
- **Replay is strategy-free.** replay() just folds the logged actions; it never instantiates a strategy. This is what makes an external, non-deterministic strategy safe.
- **Never trust a strategy.** The engine validates every decision with validateFormation. An illegal or late decision is replaced by the standard built-in's decision and the substitution is logged (strategyFallback { reason }).
- **Client cannot forge them.** Stranger actions are accepted only through *internal* Convex functions; the public applyAction validator rejects them (otherwise a player could choose their own opponent's moves).
- **Built-ins run in-line.** Built-in strategies are synchronous and pure; the driver loop runs inside the same mutation that applied the player's action, so a player action and the stranger replies commit atomically.

5.3 Exact evaluation

A match outcome depends only on two d6 and known totals, so a match can be evaluated **exactly** (36 outcomes):

```
evaluate.matchOutcomes(match): { pWin: { party: number; strangers: number }; pTie: number;
  pStalemate: number }
evaluate.expectedLoss(match, side, valueOf): number // casualty probability × value, incl.
  the 2-front-line casualty die
evaluate.formation(formation, valueOf): { expectedDiff: number; breakdown: PerMatch[] }
```

valueOf(creature) is pluggable (default: victory points; alternatives: strength, replacement cost).

5.4 Built-in levels and the baseline objective

“Maximum effect” is undefined in the requirement (gap G1). Proposed definition, to be confirmed:

Objective (one round): maximise $E[\text{party value lost}] - E[\text{stranger value lost}]$,
ties broken by total effective strength, subject to the legality layer.

The dice roll is per match, not per party. Peter’s rule is that a strength gap above 5 in a match decides it in advance. So the evaluator clamps each match’s margin to ± 5 (or uses its win probability), and a round in which every achievable pairing is already decided skips the search and goes to the standard heuristic. Reasoning and its effect on the learned strategy: [neural-net feasibility study](#) §2.6 and §4.3a.

Level	Behaviour	Intent
novice	Legal but deliberately naive: runs only part of the default loadout (e.g. skips the Shield and Ring), pairs by simple strength ranking, never uses the Shield against casters	Easier game for new players
standard	The default loadout (§4.4.1) plus greedy pairing on the objective; handles Spectre/Demon/Shield/Staff correctly; $O(n^2)$	Default; also the universal fallback
expert	Optimises the artefact assignment too (it must beat the default loadout in the bench), then exhaustive or branch-and-bound over formations using evaluate, with a node budget and the standard answer as the incumbent; one-round lookahead for persistence	Hard mode; must fit within the budget below

expert must be **budgeted**: Convex mutations have bounded execution time and read/write limits (the project’s `convex/_generated/ai/guidelines.md` documents the transaction limits; the CPU budget must

be verified against current limits before sizing the search). If the budget is exhausted it returns the best incumbent — never an error.

5.5 Registry and difficulty

- StrangerStrategy instances register in `strategy/registry.ts` by id.
- The level is chosen at game creation and stored in `variants.strangerLevel` (immutable for the game, like `extensionKit`). Leaderboards are segmented by level (G7).
- Test Mode gains a “force strategy / script stranger formation” control (G9).

5.6 External (neural-network) strategy

A Convex **action** (actions may fetch; mutations may not) orchestrates it:

1. Client calls a public action (e.g. `game.applyActionAsync`), which loads the state via a query.
2. For each stranger decision needed it builds a **minimised view** (only what a human sees) and fetches the model endpoint, with a timeout and bounded retries.
3. It then calls an **internal mutation** (`ctx.runMutation(internal.game.applyStrangerDecision, ...)`) with the player’s action and the decisions, which reduces and logs them atomically.
4. On timeout, error or an illegal answer it uses the standard built-in and logs `strategyFallback`.

Contract items still needed (G2): endpoint and auth (secret via `convex env`), request/response schema (versioned), latency SLA and timeout, cost ceiling, where training data comes from (the harness in §8 can generate it), and whether the model may see anything beyond the human-visible view.

6. Server design (apps/web/convex)

Change	Detail
Variants	Add <code>combatRevision?: boolean</code> and <code>strangerLevel?: "novice" "standard" "expert" "nn"</code> to the variants object in <code>schema.ts</code> ; thread through <code>newGame</code> , <code>startTestGame</code> , <code>replay/replayByCode</code> , <code>highScores.log</code> , <code>ReplayBundle</code> , and <code>MpGameState.variants</code> . The engine’s <code>replay(seed, picks, actions, variants, testMode)</code> variants parameter gains the same keys.
Action validators	Add the new actions to both <code>game.ts</code> and <code>multiplayer.ts</code> validators. Improvement I9: define the action schema once (in the engine) and derive both validators so they cannot drift.
Stranger actions	Registered only for internal functions; rejected by the public <code>applyAction</code> .
Driver	<code>applyAction</code> / <code>multiplayer</code> equivalent call the built-in driver loop after reducing; external levels use the action path in §5.6.

Change	Detail
Log	gameEvents already stores action: <code>v.any()</code> ; stranger decisions fit without a schema change. Add <code>strategyId/strategyVersion</code> fields inside the action payload. Old logs remain valid for flag-off games, which keep the old engine. Not enough for training data or for replaying across engine versions: see §6.1.
Saved games	A <code>combatRevision: false</code> game keeps <code>strangers: number[]</code> and the old engine. Enabling the flag on an existing game is not supported (a fight cannot change rules mid-game); only new games opt in.
Document size	Stranger instances and persisted matches add a few hundred bytes to state; far below the 1 MB document limit.

6.1 Log content: what a recorded game must carry

A review of the current log (`seed + picks + variants + testMode`, then `{action, events}` per move) found it replays a game exactly but cannot yet serve as training data, nor survive the rules change:

- **Engine/rules version.** The log has only a format version: 1. Replaying an old log through the revised engine diverges (the dice are consumed differently), so each log records the engine/rules version and the `combatRevision` flag, and old logs are replayed only by a pinned engine.
- **Structured combat record.** `combatRoll` holds display strings (“W-Hero + Man”) and totals. Add a `fightRoundResolved` event per match with creature ids and borne artefacts for both sides, each side’s effective strength with its breakdown, the formation chosen, who attacked, and which strategy decided. Training rows are then self-contained and need no replay.
- **Stranger decisions.** Loadout and pairing are already logged as tagged actions (§5.2); they are the only labels a learned strategy can be compared with, because today only the *player’s* pairing is recorded.
- **Multiplayer.** Only the solo path appears to write `gameEvents` (multiplayer logs narration messages), so multiplayer fights have no per-seat action log. To be confirmed, then fixed with a seat on every recorded action.
- **Test-mode games** carry `testMode` and must be excluded from training data.

7. Front-end design (R5)

7.1 Step-wise fight surface

`FightSurface` becomes a thin renderer over `fight.pending`:

Pending	Player sees	Interaction
<code>scroll</code> (party attacking)	Scroll prompt with the three variants and the curse warning	Confirm-gated (existing <code>ConfirmButton</code>)

Pending	Player sees	Interaction
deploy (party defending)	“The strangers attack — choose your front line” with the minimum shown (“deploy at least 3”)	Pick creatures; Done enabled only when legal
engage (party attacking)	Strangers’ deployed line; pick a party creature then a target	Targets limited to legalEngagements
redploy (party larger)	Remaining free creatures and background; slot hints	Drop targets limited to legalPlacements
strangers deciding	A short, skippable narration of the strangers’ choice (“The Ogre squares up to your Wizard — the Shield cancels its magic”), from the decision’s why trace	Read-only
casualty	Existing casualty prompt; for stranger casualties, a nomination prompt (Q-R5)	As today

7.2 Making illegal formations unrepresentable

- place / unplace in `fightPlan.ts` are replaced by `useFormation(state, side)` which wraps `legalPlacements / completeFormation`. A creature’s illegal drop targets are **disabled**, not rejected after the fact; a caster cannot be dropped behind an empty front line; “Done” is enabled only when `validateFormation` is ok.
- REASON, the Spectre/Demon constants and the forced-round derivations are deleted from the component; the engine returns obligations and plain-language reasons.
- **Improvements:** a *Suggest* button that fills the formation from the standard strategy (the same code the strangers use — a free hint system for new players); per-match win odds from `evaluate` (exact, no simulation); “mandatory” markers on creatures that must deploy.

7.3 Everything else the surface touches

Celebrations (`dragonSlain`, `sorcererSlain`) are unchanged. Stranger relics render with the existing `FightCard` relic corner. The roster and `PartyPanel` need no change except that **stranger** artefacts are now visible on the fight surface. `PvpFightSurface` follows in M7.

8. Testing and strategy evaluation at scale

8.1 Layers

Layer	What	Tooling
Unit	strength, artefacts, round phases, casualty, legality against worked examples from Peter’s documents (the Hero-	vitest

Layer	What	Tooling
	vs-Priest/Troll/Man/Dwarf example, the Ogre/Troll example in the consolidated rules)	
Property	For random scenarios: every legal-move enumeration yields formations <code>validateFormation</code> accepts; <code>completeFormation</code> always valid; engine never panics on any strategy output; round conservation (nobody both dies and survives); alternation invariant	vitest + a property library (propose fast-check; none is installed today)
Golden	New narrative goldens for <code>combatRevision: true</code> fixtures; existing goldens must not change with the flag off (this is the safety net, as the solo/kit/multi golden firewalls already work)	existing snapshot tests
Strategy bench	§8.2	new CLI

8.2 Strategy bench harness (`packages/engine/src/combat/strategy/bench`)

Goal: confirm and compare strategies *exhaustively*, as the requirement asks.

- **Scenario space.** A scenario is (party multiset + artefacts, stranger multiset + artefacts, level, surprise, who attacks). Enumerate by **canonical multisets** (creatures of the same type are interchangeable up to artefact placement), which cuts the space by orders of magnitude. A chamber draw is bounded by `MAX_STRANGERS = 8` (at most 6 cards in practice), but a **Mutiny** can make the stranger side far larger (ceilings 37 / 48, feasibility study §2.6), so Mutiny-sized fights are a separate stratum: the bench samples them and the standard heuristic decides them. The exhaustive sweep targets small fights (≤ 4 per side) and *samples* above that. (Estimated counts must be measured — they are not asserted here.)
- **Exact round evaluation.** Because a match is two d6, the bench scores a strategy's formation with `evaluate.formation` (no dice, no variance) — so strategies can be ranked on millions of scenarios cheaply.
- **Full-fight arena.** For multi-round behaviour (persistence, alternation), seeded Monte Carlo plays fights to the end: `strategyA` (strangers) against a fixed set of **reference party policies** (greedy, cautious, magic-heavy, random-legal). Metrics: stranger win rate, mean party losses, mean rounds, artefact-use rate, and the regret against expert.
- **Legality sweep.** Every strategy decision across the whole space is run through `validateFormation`; a single illegal output fails the bench.
- **Reproducibility.** Each run is (`strategyId`, `version`, `scenario set hash`, `seed`) → a JSON report; reports are diffable for regression gating.

- **Budgets.** expert is measured for time and node use per decision; the external strategy is excluded from exhaustive runs (replaced by a recorded-decision replay).
 - **CLI.** `pnpm --filter @sorcerers-cave/engine bench:strategies --levels standard,expert --max-side 4 --seed 1`. A small subset runs in CI; the full sweep runs nightly.
 - **Difficulty calibration.** The bench provides the evidence for what novice/standard/expert mean (target win-rate bands), which the requirement leaves undefined (G1, G10).
-

9. Questions, gaps and improvements

9.1 Questions for Peter (rules)

- **Q-R1** — *Triggering a Fight* is a named step in the overview but neither document defines it. Who attacks in round 1 for each start — hostile reaction (strangers, surprise), party Attack, pacified-chamber Attack, Demon ambush, retreat bounce-back (§4.2.0) — and does the role strictly alternate every round in solitaire, including when the party would rather keep attacking?
 - **Proposed default:** Strangers attack on hostile/Demon; party on Attack; strict alternation
- **Q-R2** — Scroll (your own note): is the “destroy” variant usable in *any* round the party attacks, with the two surprise variants only before round 1?
 - **Proposed default:** Destroy: any attacking round; surprise variants: round 1 only
- **Q-R3** — “Front line of any match does not exceed two” — per side, with 2v2 still forbidden (as in `validatePlan` today)? Can unengaged creatures of the larger side join an *already engaged* match as the second front-liner?
 - **Proposed default:** Per side; 2v2 banned; joining allowed
- **Q-R4** — When do magic users commit to the background for each side? Step (b)/(c) mention only front-line deployment; the PvP flow has a separate “defender casters” step. Can any number of casters back one match, and may a lone caster stay in the background?
 - **Proposed default:** Casters placed in step (d) for the larger side and with the front line for the smaller; lone caster must fight
- **Q-R5** — Stranger casualty: the winning player nominates and on 4–6 the **other** dies — is the nominated creature the one **spared**? Whose Ring adds the +1?
 - **Proposed default:** Nominated = spared; losing side’s Ring adds +1
- **Q-R6** — Ring bonus: must the bearer be among the *fighters* (front/background) for the side to get +1, or is any living holder enough (today’s behaviour)?
 - **Proposed default:** Any living holder (current behaviour)
- **Q-R7** — *Partly answered:* the default loadout says the pool is the artefacts **present** and leftovers stay on the floor. Still open: is it (re)run only at fight start, or also when a bearer dies, when Mutiny deserters join, or when strangers arrive later (a Demon)? May strangers pick up artefacts dropped mid-fight?
 - **Proposed default:** Run at fight start and again at the next deployment after a bearer dies; no mid-fight pickup
- **Q-R8** — The *WHO CAN USE?* table says who may bear; the cards say who gains a bonus (Sword: Man/Woman/Hero only). Confirm bonuses follow the card text, so a Priest-class stranger with the Sword gains nothing. Also an anomaly: the Ring card says a DWARF may wear it, and the table allows a Dwarf **stranger** but not a Dwarf **ally** — typo or intended?
 - **Proposed default:** Bonus per card text; treat the Dwarf-ally Ring blank as a typo unless told otherwise

- **Q-R9** — Where do Spectre/Demon sit in the new flow (they cannot be fought hand-to-hand)? When does “the strongest creature is automatically slain” occur — before step (e)? Can a Spectre be a stranger *attacker’s* paired creature?
 - **Proposed default:** Resolved at step (e) before matches; Spectre/Demon are “engageable only by magic/Sword/Axe/Shield”
- **Q-R10** — During a round where the strangers attack, may the player use artefacts (Strength Potion, Lotus Dust, Holy Water) or only deploy?
 - **Proposed default:** Only deploy; consumables on the player’s own attacking round
- **Q-R11** — Hidden Cards variant: may strategies see face-down player cards?
 - **Proposed default:** No (human-visible only)
- **Q-R12** — “Defender must attempt to deploy at least as many creatures”: do casters count as deployable fighters?
 - **Proposed default:** Yes — a caster may fight hand-to-hand
- **Q-R13** — Multiplayer: when several parties or a union face strangers, who is attacker/defender and in what order do they act?
 - **Proposed default:** One party at a time, as today
- **Q-R14** — *Ending a Fight* is referenced by Peter’s step (f) and the overview, but **no document defines it**: end conditions (including “put to sleep”, e.g. Flute-lulled Dragons), the retreat procedure and timing, what happens to engaged creatures and loot. Also confirm that *Wrap Up the Round* and *Ending a Fight* are round/fight-level, not per-match.
 - **Proposed default:** Today’s behaviour (§4.2.0) until a document arrives
- **Q-R15** — Scope and timing of *Stranger Fight Preparations* (*stranger party only*): is it only the artefact loadout, or also an initial stance (who starts in the background)? Does it run once before round 1 only, or again for reinforcements (Mutiny deserters, a Demon arriving later)?
 - **Proposed default:** Loadout only; once before round 1, and again at the next deployment after a bearer dies or strangers are added
- **Q-R16** — (*New.*) The loadout document defines how strangers are **equipped**; is there a matching default **pairing** procedure for deploy / engage / redeploy / background, or should one be derived from the G1 objective?
 - **Proposed default:** Derive from the G1 objective until supplied

9.1a Questions and discrepancies raised by the default loadout

Checked against the consolidated rules and the engine (combat.ts, effects.ts, data/creatures.ts):

- ✓ **Q-L1** — *ANSWERED, see §9.1b. Thief.* The Sword and Axe lists put the Thief above the Woman, but only Man/Woman/Hero (and Dwarf for the Axe) gain a strength bonus — a Thief gains **nothing**. For the **Shield** the engine only lets Man/Woman/Hero/W-Hero use it (SHIELD_WARD_ELIGIBLE), so a Thief-borne Shield is **inert**.
 - **Question / proposed default:** Intended (ordering by “able to bear”)? Or should lists follow *who gains*, and should the Thief be dropped from the Shield list? Default: keep the lists as written, but skip an artefact for a bearer who gains nothing from it.
- ✓ **Q-L2** — *ANSWERED, see §9.1b. Staff.* The list prefers Sorcerer, Apprentice, Wizard, Witch, Priest, Scholar. The engine only adds the Staff bonus for Priest (+1) and Wizard (+2); the card says “+2 for a Priest or Wizard”. So Sorcerer/Apprentice/Witch/Scholar gain nothing today, while the list ranks them above the Wizard. (The Apprentice “uses artefacts as a Wizard”, usesArtifactsAs, but casterMP does not apply it.)
 - **Question / proposed default:** Does the Staff bonus apply to the whole Priest class? At what values (card +2 vs engine Priest +1)?

- ✓ **Q-L3** — *ANSWERED, see §9.1b. “Weapon”*. Used in the Axe (“with no weapon”) and Shield (“with a weapon”) lists and in Step 5 (“weapon artefacts already borne”).
 - **Question / proposed default:** Sword only, or Sword and Axe (the card says only one may be used per fight)? Is the Staff a weapon for Step 5? Default: weapon = Sword or Axe.
- ✓ **Q-L4** — *ANSWERED, see §9.1b. Ring “adjusted strength”*. Step 5 allocates to the highest adjusted strength.
 - **Question / proposed default:** Fighting strength only, or total (FS + MAG), including the Staff bonus? And must the Ring-bearer actually be a fighter (this is Q-R6; if yes, the strategy must guarantee the bearer is engaged).
- ✓ **Q-L5** — *ANSWERED, see §9.1b. “Eye present in the area”*. The engine’s eyeActive is true only when a *living party member holds* the Eye. An Eye lying in the chamber has no effect today. The card says artefacts are powerless “when this gem is in the same area”.
 - **Question / proposed default:** Should eyePresent = held by the party **or** on the floor in the area (affecting both sides’ artefacts)? Default: yes.
- **Q-L6** — **When it runs**. The document defines the allocation, not the trigger (see Q-R7). It also says “creature in the party” in Step 5 (presumably the *stranger* party).
 - **Question / proposed default:** Confirm the trigger and the wording.
- ✓ **D1** — *ANSWERED, see §9.1b. Data:* the rules list **Hero STR 6**; the engine has Hero fs 5 (cost 6). Hero strength decides the Ring allocation and every worked example.
 - **Question / proposed default:** Which is correct?
- ✓ **D2** — *ANSWERED, see §9.1b. Data:* the rules list **Spectre STR 4 (MAG 4)**; the engine has fs 0, mp 5.
 - **Question / proposed default:** Which is correct?
- **D3** — **Data:** the Staff card says +2 for Priest **or** Wizard; the engine gives the Priest +1 (combat.ts, SC-9.3-4).
 - **Question / proposed default:** Is Priest +1 a deliberate house rule?

All other creature strengths I compared (Apprentice, Demon, Dragon, Dwarf, Giant, Lion, Man, Ogre, Priest, Scholar, Sorcerer, Thief, Troll, Unicorn, Witch, Wizard, Wolf, Woman, Woman-Hero) agree between the rules text and the engine.

9.1b Peter’s response (03-OCT-2026) and what it changes

Source: docs/rules/response-to-feedback-20261003.md, with the revised loadout strangers-default-loadout-v2-20261003.md and the revised Consolidated-Rules-20260925-2.pdf.

Item	Peter’s answer	Effect on this specification
Q-L5 Eye	Always on in the area; need not be borne (Talisman wording contrasted)	eyePresent = held or on the area floor, both sides (§4.4.3); loadout skip rule
Q-L1 Thief	Error in the list, “though not for the reason you gave”: a Thief uses artefacts as a Man, so it <i>can</i> use Sword/Axe/Shield. Woman outranks Thief because the Unicorn rule makes her ability useful to strangers	Preference lists reordered (Woman before Thief, Sword, Axe, Shield and Ring); capability table (§4.4.2)

Item	Peter's answer	Effect on this specification
Systemic	"Uses artefacts as" and "has all the capabilities of" are not implemented correctly	§4.4.2 audit and M0 work item
Q-L2 / Sorcerer	Witch/Scholar as Priest, Apprentice as Wizard; Sorcerer as Wizard (house rule, argued from the Apprentice and the Sorcerer's power)	Capability table; Q-L7 answered: Addendum note #4
Q-L3 "weapon"	Term removed; lists reworded	§4.4.1 table; Appendix C vectors re-derived
Q-L4 adjusted strength	Strength + bonus from Sword, Axe or Staff	§4.4.1 step 5
D1 / D2	Hero 5, Spectre 5 (5); the old text was wrong	Engine already agrees; nothing to change
Loadout V2	Axe "not bearing Magic Sword"; Shield "bearing Sword or Axe"; Eye wording	§4.4.1

Still open from §9.1a: **Q-L6** (trigger and "creature in the party"), **D3** (Staff +1 vs +2), and the new **Q-L8** (Woman-Hero union reading). **Q-L7** was answered by Mark on 04-OCT-2026 (Addendum).

9.2 Technical questions

- **Q-T1** — Is `combatRevision` a per-game variant that defaults off, or do we intend to migrate everything and retire the old engine? (This spec assumes per-game, with a later flip of the default.)
- **Q-T2** — Hosting, auth, latency SLA and cost ceiling for the neural network; may we send full state or only the human-visible view?
- **Q-T3** — What CPU budget can expert use inside a mutation? (Needs a measurement against current Convex limits.)
- **Q-T4** — Should PvP migrate onto the new engine (M7), or stay on `multi-fight.ts`?
- **Q-T5** — May we add a property-testing dependency (`fast-check`)?

9.3 Gaps in the high-level specification

- **G1** — "Maximum effect" and the difficulty levels are undefined. §5.4 proposes an objective and level behaviours; they need a product decision.
- **G2** — The neural-network strategy has no contract: model, inputs, SLA, hosting, cost, fallback, privacy, versioning, ownership.
- **G3** — No statement on **information available to strategies** (hidden cards, other parties' secrets).

- **G4 — Backwards compatibility** is unaddressed: in-flight games, saved codes, replays and golden snapshots all depend on the old rules.
- **G5** — Multiplayer scope is unclear: PvE only, or PvP too? Timers (45 s windows) vs stranger decisions?
- **G6** — The UX for *defending* and for step (d) is new and needs wireframes (including mobile and accessibility).
- **G7** — Difficulty changes outcomes, so **leaderboards must be segmented by level** (they are already segmented by kit).
- **G8** — Logging and replay for stranger decisions, and how a bug report surfaces “why the strangers did that”.
- **G9** — Test Mode support (force a level, script a stranger formation) for QA.
- **G10** — Success metrics: target win-rate bands per level, decision latency, fallback rate.
- **G11** — Ordering: R5 (UI legality) is independent of R1–R4 and could ship first on the current engine.
- **G12** — Rules authority: Peter’s documents vs the engine spec — the new rows must be traceable to his steps (Appendix A).
- **G13** — Two steps of the new overview have **no rules document**: *Triggering a Fight* and *Ending a Fight*. The overview’s step names and Peter’s step names also differ (“Match is Resolved” vs Step iv); §4.2.0 maps them.

9.4 Proposed improvements

ID	Improvement
I1	Engine-owned legalMoves / completeFormation, consumed by UI and strategies (§4.5).
I2	A <i>Suggest formation</i> hint for players, using the standard strategy (§7.2).
I3	Exact win-probability display per match from the evaluator.
I4	Every strategy decision carries a short why trace — drives the narration, bug reports and the bench.
I5	Strategy id/version/seed recorded in every logged stranger action.
I6	Difficulty-aware leaderboards.
I7	Bench in CI plus a strategy regression ladder (a new strategy must not lose to its predecessor).
I8	

ID	Improvement
	Delete the rule constants duplicated in <code>FightSurface</code> ; one source in the engine.
I9	Define the action schema once and derive the two <code>Convex</code> validators from it.
I10	Unify PvP and PvE on the round engine; retire <code>multi-fight.ts</code> pairing code.
I11	Remove dead state (<code>FightState.focus</code> , <code>strongestStranger</code>) if confirmed unused after the rewrite.
I12	Quick win now: ship the <code>legalPlacements</code> UI guard for the current <code>BattlePlan</code> model (R5) before the engine work.
I13	One <code>capabilities</code> table (“uses artefacts as” / “capabilities of”) used by loadout, strength, Staff, Shield ward and the Spectre/Demon predicates; replaces hard-coded id lists (§4.4.2).
I14	<code>eyePresent(state)</code> — Eye in the party or on the area floor — replacing <code>eyeActive</code> for fights (§4.4.3).

10. Delivery plan, risks and spec maintenance

10.1 Milestones

M	Deliverable	Exit criterion
M0	Rules sign-off (Q-R1–R15, Q-L6, Q-L8) including the missing <i>Triggering</i> and <i>Ending a Fight</i> definitions (G13), resolve the data discrepancy D3 (D1, D2 closed), capability table (I13) and area-wide Eye (I14), decisions on G1/G2/Q-T1, fixtures from Peter’s worked examples and the loadout vectors (Appendix C)	Written answers; fixtures committed; rules and engine data agree
M1	<code>model</code> , artefacts, strength, legality behind <code>combatRevision</code> (no flow change); unit + property tests	All existing tests and goldens unchanged; new suites green
M2	round engine, stranger inventory, persistent matches,	

M	Deliverable	Exit criterion
	new actions, standard strategy; driveable from a CLI	A full fight plays to the end headlessly with the flag on; new goldens
M3	Server wiring: variants, validators, internal stranger actions, in-line driver, log/replay	Replay of a flag-on game reproduces exactly; public API cannot forge stranger actions
M4	Step-wise FightSurface + useFormation (R5), narration, Suggest, odds	No illegal formation can be built; UI tests; manual playthrough
M5	novice/expert, bench harness, calibration report	Bands agreed (G10); legality sweep clean
M6	External strategy: Convex action, recording, fallback, secrets	NN decisions logged and replayable offline; fallback rate measured
M7	PvP alignment (Q-T4), default flip (Q-T1), docs	Decision-dependent
M4a (optional, earlier)	I12: legalPlacements guard on the current plan model	R5 delivered without engine change

10.2 Risks

Risk	Mitigation
Rules ambiguity (Q-R, Q-L) produces rework	M0 gate; fixtures encode answers
The engine disagrees with the rules on who can use artefacts and on the Eye (§4.4.2, §4.4.3), and on the Staff bonus (D3), so tests and strategies encode the wrong behaviour	Fix both in M0, flag-gated; calibrate strategies only afterwards
strangers: number[] → instances touches reactions, hazards, Scroll, Holy Water, statues, rendering	Accessor strangerIds(state) first; flag-gated; golden firewalls
Persistent engagements break saved games	Per-game flag; no mid-game switching
expert exceeds Convex execution limits	Node/time budget with the standard incumbent; measure first (Q-T3)
External strategy latency or outage	Timeout + fallback + recorded decisions (§5.6)
Hand-maintained validators drift again	Single schema (I9); regression test
Two fight UIs diverge	M7 alignment; shared useFormation

10.3 Spec maintenance (CLAUDE.md)

Every engine change in M1–M3 updates docs/specs/engine-spec.md in the same change. The new requirement rows are SC-CMB-* and **supersede** the corresponding rows when `combatRevision` is on: SC-9.1-* (pairing and legality), SC-9.2-* (surprise), SC-9.3-* (strength, ring, curses, match resolution), SC-9.4-* (Spectre/Demon/Dragon/Sorcerer/Eye), SC-9.5-* (casualties, retreat, flow). The legacy rows stay, marked *applies when combatRevision is off*.

Appendix A – Traceability: Peter’s steps to the design

Peter	Rule	Design element	Test
Fighting a Match, Step i.1	Magic users in background	<code>strength.effectiveStrength</code> positional effect	worked example “Hero vs Priest/ Troll/Man/Dwarf” (= 9)
Step i.2	Eye, Axe, Shield, Staff, Sword, Talisman, Ring	<code>artefacts.ts</code> rules + <code>MatchContext</code>	per-artefact table tests, incl. Eye nullification
Step i.3	Spectre, Demon	<code>strength</code> immaterial rules; legality engageability	forced-slay and Shield-standoff cases
Step ii	Total effective strength	<code>sideTotal</code>	sum properties
Step iii	Result = TES + d6 + surprise{0,1,2} + ring{0,1}	<code>result</code>	exact <code>evaluate</code> vs enumeration
Step iv	Stalemate / tie / win; ties unresolved	outcome; persistence in round	persistence tests
Step iv (casualty)	1 front → dies; 2 front → nomination + d6	<code>casualty.ts</code>	party and stranger variants (Q-R5)
Each Round, Step a	Scroll	round phase a; <code>artefacts</code> Scroll rule	attacker-only, Eye, three variants
Step b	Defender deploys	<code>legalDeployments</code> ; <code>deploy</code> action	minimum-deploy property
Step c	Attacker engages	<code>legalEngagements</code> ; <code>engage</code> action	one-to-one property
Step d	Finalise deployment	<code>legalPlacements</code> ; <code>redeploy</code> action	match-count and ≤2 properties
Step e	Fight each match	round phase e	goldens
Step f	Wrap up; alternate		

Peter	Rule	Design element	Test
		round phase f; fight.attacker flip	alternation invariant
Overview	Triggering a Fight	FightTrigger; trigger phase	every start path sets attacker and surprise
Overview / strangers- default-loadout- v2-20261003.md	Stranger Fight Preparations	prep phase; defaultLoadout	Appendix C vectors V1-V15
Overview (referenced by step f)	Ending a Fight	end phase (provisional = SC-9.5-6...11)	wipe-out, clear-out, retreat, dead-end bounce-back

Appendix B – Worked example (strangers attack, round 2)

State: round 1 was the party's (Ogre-vs-Hero match unresolved after a tie). Round 2: **attacker = strangers**.

1. (a) Strangers cannot read the Scroll — skipped.
2. (b) The player is the **defender**: the strangers have 2 unengaged creatures (Troll, Priest); the party has 3 unengaged (Woman, Dwarf, Wizard). The player must deploy ≥ 2 ; deploys Woman and Dwarf (Wizard held back).
3. (c) The strangers' engage decision (strategy standard) pairs Troll $\rightarrow$ Woman and Priest $\rightarrow$ Dwarf (the Priest bears the Magic Staff: MAG 2 + 1). Two new matches are created; Ogre-vs-Hero persists.
4. (d) The larger side (the party, 3 vs 2) may redeploy its unengaged creatures — the Wizard may join the Woman as a second front-liner (2v1, legal) or sit in the background behind the Dwarf.
5. (e) Three matches are fought. Totals use the **same** effectiveStrength for both sides; the stranger-borne Staff is applied. Results: one win, one tie (persists), one loss.
6. (f) The dead match is dissolved; backgrounds become unengaged; **attacker flips to the party**.

All three stranger decisions (assignArtefacts at fight start, engage here) appear in the log as actions tagged with strategyId, and replay reproduces the round without running a strategy.

Appendix C – Test vectors for the default loadout

defaultLoadout is deterministic except for the random tie-break; every vector below avoids ties unless stated. Strengths are the engine's, which now agree with the revised rules (D1, D2 closed).

#	Strangers	Artefacts present	Eye	Expected allocation
V1	Hero, Man, Dwarf	Sword, Axe, Shield, Ring	no	Sword $\rightarrow$ Hero · Axe $\rightarrow$ Dwarf (first preference) · Shield $\rightarrow$ Hero (bearing the Sword) · Ring $\rightarrow$

#	Strangers	Artefacts present	Eye	Expected allocation
				Hero (adjusted $5+2 = 7$ beats Dwarf $1+3 = 4$, Man 3)
V2	Hero, Man	Sword, Ring	yes	Nothing allocated; both stay on the floor
V3	Man, Woman, Thief	Sword	no	Sword → Man (3rd) before Woman (4th) and Thief (5th)
V4	Wizard, Sorcerer, Priest	Staff, Ring	no	Staff → Sorcerer (1st) · Ring → Sorcerer (adjusted $13+2 = 15$ with the Staff, beats Wizard 7 and Priest 4)
V5	Hero, Man	Sword, Axe	no	Sword → Hero · Axe → Man (“Hero not bearing Magic Sword” is ineligible: Hero bears it)
V6	Man, Woman	Shield	no	Neither bears a Sword or Axe, so the “bearing neither” half applies: Man (8th) before Woman (9th) → Man
V7	Ogre, Troll	Sword, Ring	no	Both are inhuman and cannot bear either; both stay on the floor
V8	Man, Man	Sword	no	Exactly one Man bears it, chosen by the decision substream; for a fixed seed the

#	Strangers	Artefacts present	Eye	Expected allocation
				result is repeatable, and across seeds both outcomes occur
V9	Hero	Sword, Axe, Shield, Ring	no	Sword → Hero · Axe: “Hero not bearing Magic Sword” fails, nobody else → stays on the floor · Shield → Hero · Ring → Hero
V10	Thief, Woman	Sword	no	Sword → Woman (4th) before Thief (5th) — V2 of the document
V11	Thief	Sword, Axe, Shield	no	Thief counts as a Man (§4.4.2): Sword → Thief · Axe: “not bearing Magic Sword” fails, stays on the floor · Shield → Thief (bearing the Sword)
V12	Witch, Scholar	Staff	no	Both count as a Priest; Staff → Witch (4th) before Scholar (6th)
V13	Apprentice, Priest	Staff, Ring	no	Staff → Apprentice (2nd, counts as a Wizard) · Ring → Apprentice (adjusted 9+2 = 11 beats Priest 4)
V14	Hero, Man	Sword	yes, lying on the floor, not held by anyone	Nothing allocated (Q-L5, §4.4.3)
V15	Woman-Hero, Man	Sword	no	Sword → Woman-Hero (2nd) before

#	Strangers	Artefacts present	Eye	Expected allocation
				Man (3rd); she gains what a Hero would (+2)

The vectors follow V2 of the loadout document and Peter's capability rulings (§4.4.2). V12 and V13 need the capability table; V4's Sorcerer rows depend on the house rule (Addendum note #4). Ties are not asserted in any vector except V8.